

A Hybrid Approach for Efficient and Precise Food Detection using YOLOv9 and RCNN

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Abstract. Identifying food items on plates is difficult. There is a wide variety of foods. Each food item can be complex. Because different backgrounds on the plates also create difficulties. Additionally, food items can be occluded by other objects. These factors make detection more complex. In this paper, a hybrid model is proposed to overcome such problems. The hybridization of two state-of-the-art deep learning models: Mask R-CNN and YOLOv9 are used for food detection on plates. These models are evaluated against a broad set of plate images based on metrics including mean average precision (mAP), precision, recall and inference speed. The result shows the significant improvement that the hybrid model performs better, with a score of 91.57% mAP and a 90.17% F1 score. The result is better than other state of art which successfully balancing speed and accuracy in detection and segmentation in complex scenarios. After analyzing the proposed model the potential suggestions for future directions are suggested, such as handling occlusions and increasing robustness for small objects.

Keywords: Mask R-CNN, YOLOv9, Convolutional Neural Networks (CNN), Detection, Instance Segmentation, Image Segmentation.

1 Introduction

The detection of food is an important task for several applications, such as food safety, dietary analysis, and also robotic manipulation. The ability to accurately and rapidly detect food enables automated systems to inspect food quality, help with serving size, and perform smart food preparation [1]. Despite this, identifying food items on plates as shown in **Fig. 1**. has its own difficulties because of the composition diversity of the food, the diversity in plate sizes and backgrounds, air background, and occlusion objects.

Food detection based on traditional methods typically relies on the segmentation of food objects based on color, their shape analysis, or hand-crafted features [2]. These

methods achieved some improvements but are limited because of the sensitivity of lighting conditions, object occlusions, and complexity of food items.

Deep learning has been proven to be more efficient than traditional methods for object detection tasks in the past few years. Convolutional Neural Networks (CNNs) [3] is another successful architecture used to solve food detection.



Fig. 2. Some examples of Indian platters dishes

The motivation behind this work lies in addressing challenges in automated food detection, such as overlapping objects and variable lighting. Reports show the global food technology market is growing at a 6.1% CAGR, highlighting the need for innovations. Additionally, 30-40% of global food waste could be reduced with better detection systems. This research assesses the modeling capabilities of the two cutting-edge deep learning food detector algorithms, Mask R-CNN [4] and YOLOv9 [5], on plate food detection. Mask R-CNN excels in its accuracy for object detection and segmentation; YOLOv9 is known for its speed and efficiency. A comparative analysis will be performed on a diverse set of plate images to identify the most suitable model for real-world applications.

2 Literature Review

Intelligent object detection in food samples is obtained with the help of viable machine learning models to identify, classify and localize objects in images [6]. Early methods for object detection were based on handcrafted rule methods, which took a lot of time and could not be scaled properly. Since Deep learning has become popular. Convolutional neural networks (CNNs) are now widely used. They enable quick and accurate detections. CNNs are fast and reliable for many applications. They can analyze food samples, as shown by Alahmari, S. S., et al., [7].

Ren, S., et al [8] proposed a Region-based convolutional neural networks (RCNN) model which is one of the most significant early milestones in this domain to achieve a high level of accuracy.

But even though the RCNN are accurate with their predictions, the time taken is really high, which leads to the situation of this not being applicable for a real-time situation. In an attempt to overcome such limitations, region proposal networks were introduced into Faster RCNN in research done by Shaoqing Ren et al and Mask RCNN with streamlined and low-latency implementations [9, 10].

Movement by the YOLO (You Only Look Once) series that allowed real-time object detection with a single-pass neural net architecture [11]. Bochkovskiy et al. (2020) found that YOLO models have low detection times and high processing speeds [12]. They suggested that these models are more suitable for real-time applications. This includes tasks like food sorting and quality inspection. The latest version, YOLOv8, is especially effective for these purposes. YOLO models are typically faster mentioned by Bochkovskiy et al [12]. However, YOLO models sacrifice some accuracy in comparison to other techniques, and can be important for detecting contaminants from whether food samples [12]. The trade-off of detection speed vs accuracy has made rounds in literature and researchers have been torn to a great deal on its optimality depending on the application requirements. For example, in contamination monitoring, where slightest errors might have catastrophic effects, high accurate but slow RCNN models are still preferred over fast but low accurate YOLO-based models [10-12].

Recently, some studies emphasize the use of hybrid models by introducing the best of both worlds in order to tackle specific challenges in object detection that cannot be easily dealt with by YOLO or RCNN alone. Those hybrid methods are able to increase model accuracy, alleviate a model's suffering from bias, and built better generalization on diverse data pairs, and that is especially crucial for the detection of food samples [13]. Combining the advantages of both models can overcome the shortcomings presented by the traditional object detectors which are required to be accurate and fast as food samples present unique challenges in the form of different shapes, textures and lighting conditions is was done by Agarwal et al. in his research paper[14-17].

Even though there has been progressing, the literature lacks the application of hybrid models specially engineered for food samples also. The majority of studies are concerned with implementing these models on the generic datasets or in the non-food environments. Moreover, there are very few works that combine the recent YOLO versions with the RCNN for the food sample. Because of the specific needs for food sample detection: the need for a reliable differentiation between possible contaminants and food particles and the diversity of samples with meaningful variability, more work is needed on hybrid models of YOLOv8 and RCNN. Agarwal et al. present a combined deep learning method designed for recognizing food items and estimating their calorie content. Their approach utilizes Mask R-CNN for accurate segmentation and YOLOv5 for effective classification. Filling up this gap may result in an improved speed as well as accuracy of object detection models in food-related applications, thus a more effective and reliable solutions in real life scenarios [13, 17].

Table 1. Comparison and key Findings

S No	Year of Publication	Key Finding and Contribution	References
1	2016	YOLO redefines object detection as regression, achieving real-time performance (45 FPS), enhanced generalization, and competitive accuracy with a unified, end-to-end trainable neural network	[11]
2	2017	Mask R-CNN extends Faster R-CNN for simultaneous object detection and segmentation, achieving state-of-the-art results on COCO tasks, with flexibility for generalizing to other tasks (e.g., pose estimation).	[10]
3	2019	The paper reviews deep learning-based object detection methods, categorizes approaches, analyzes performance, and highlights future research directions with clear visual summaries.	[13]
4	2020	YOLOv4 achieves state-of-the-art object detection accuracy (43.5% AP on MS COCO) at real-time speeds (65 FPS), introduces innovative features (Mish, SAT, Mosaic), and enables training on conventional GPUs	[12]
5	2022	The study develops IndianFood10 and IndianFood20 datasets for traditional Indian food detection, achieving a state-of-the-art mAP of 91.8% with YOLOv4, addressing challenges in boundaries and class variation.	[17]

The literature is summarized in Table 1. despite the significant progress in the design and use of object detection models in the food domain, achieving a balance of high quality and real-time performance in food sample analysis will be challenges for future directions. For complex food detection tasks, hybrid models can be an attractive and promising direction that can achieve a better trade-off between performance and efficiency. This is an area that could be explored, more specifically the integration of YOLOV8 with RCNN for improved food sample detection.

3 Methodology

Mask R-CNN [4] and YOLOv9[5] Integrated Food Detection: the Approach Mask R-CNN allows accurate instance segmentation and YOLOv9 performs fast and efficient object detection. This hybrid model takes the best of both worlds by utilizing their individual strengths to deliver accurate boundary detection and real-time inference. Unlike existing methods, it addresses challenges like overlapping objects and complex shapes by integrating speed and accuracy. The performance is measured via Recall, mAP and F1-Score: all of them assure high detection completeness, high correctness and balanced precision-recall. The effectiveness of this method surpasses that of a single model, hence making it useful for such applications like food monitoring and automatic food labelling.

3.1 The Work Flow of Proposed Model

Standard object detection metrics were used to evaluate the models, such as mAP (mean Average Precision), precision, recall, and inference speed. The models were validated on a held-out generalization test to ensure that they did not over-fit to the training data, and that they could maintain similar performance across a diversity of scenarios and conditions.

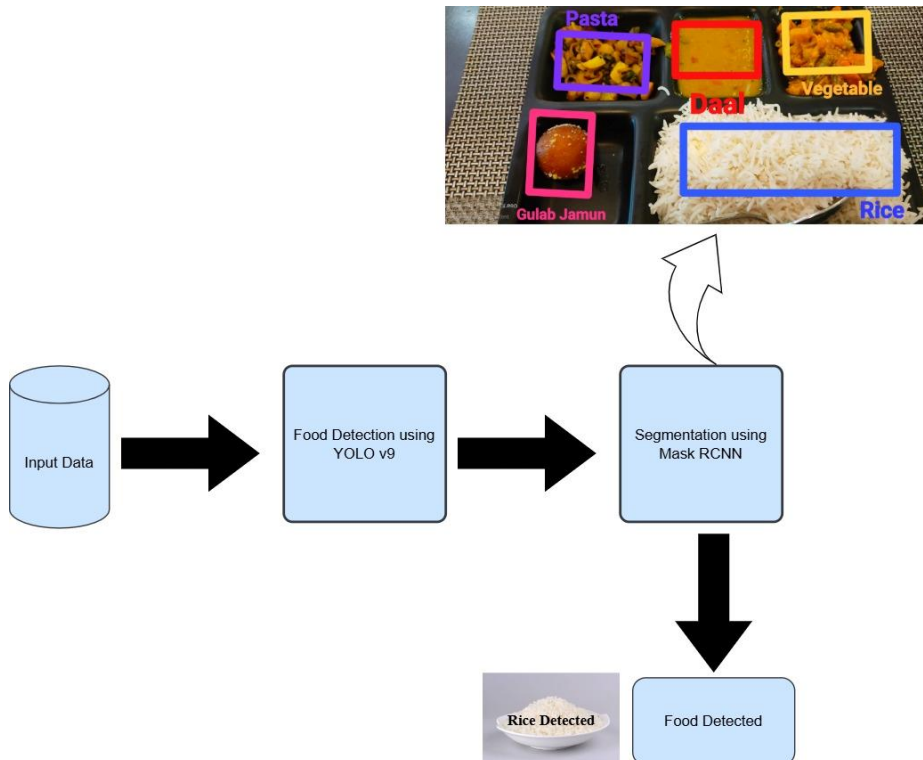


Fig. 3. The working of proposed model

A hybrid approach for food detection is shown in Fig. 2. Mask R-CNN & YOLOv9 Integrated segmentation from stacked clean masks. It starts by taking input data which basically has an image to be processed using YOLOv9. Fast in finding food items and draw bounding boxes. Fast and accurate, YOLOv9 is more suitable for real-time detection. Then the region which is detected for Mask R-CNN. This phase offers accurate segmentation looking for the exact borders of the food. These two methods are hybrid approach, making the predictions more accurate. Speed: YOLOv9, Localizing: Mask R-CNN The end product is a system which has an extremely low error rate. It is very useful for tasks such as nutritional analysis and food recognition.

3.2 Model Selection

For food detection, we choose the two state-of-the-art deep learning models: Mask R-CNN and YOLOv9.

YOLOv9: YOLOv9 is a single-stage object detection model that directly predicts bounding boxes and class probabilities from image pixels. It uses convolution to boost performance and speed.

Mask R-CNN: A two-stage object detection and instance segmentation model that extends Faster R-CNN by adding a branch for predicting segmentation masks on each Region of Interest (RoI). The first step is generating the regions with a Region Proposal Network (RPN) that classifies and refines these proposals. Lastly, it uses FCN to generate instance segmentation masks.

3.3 Dataset

For both training and testing the hybrid model, we used FoodDataset1[17]. It consists of 11,547 images belonging to 10 most popular classes of Indian Food (4 Single-dish Images and 6 Multi-dish Images). Multi-dish samples often show high variations within the same class across different images. This is because many dishes can be prepared and presented in different styles. It provides different orientations, scales and lighting conditions which is very appropriate for a real world object detection and segmentation tasks.

3.4 Model Training Approach

The hybrid model utilizes cross-entropy loss for the classification task and L1/L2 loss for bounding box regression as well as segmentation refinement. The YOLOv9 part is trained to provide bounding box predictions quickly, while the RCNN uses this info to refine detections through pixel-level segmentation. When the model is trained The Adam optimizer updates model parameters to ensure faster convergence. The overall procedure is then repeated with optimal hyperparameter tuning (for example the learning rate, batch size and number of epochs) to enhance accuracy. Lastly, they implement early stopping and decaying learning rate to avoid overfitting and improve

generalization, respectively. To maximize training efficacy, random rotation, flipping and cropping are implemented during the training process to ensure the model will generalize well across many real-world cases.

4 Results and Discussion

The experimental results point to the capability of Mask R-CNN and YOLOv9 in food detection into plates. Since Mask R-CNN provides an accurate instance segmentation masks. It can segment pixels that belong to specific food items. This segmentation is more precise, capturing the shape and extent of the food. Therefore, Mask R-CNN is better for applications that need detailed information about food shapes and sizes.

On the other hand, YOLOv9 offers a good trade-off between accuracy and speed. However, this will hurt its performance in difficult scenarios like if the food is tiny or it is hidden.

4.1 Parameter Considered

Experiments were conducted on google collab [18] and their quantitative results are summarized in Table 1. The parameter values like mean Average precision(mAP), Precision, Recall, and F1 scores are mentioned in it. This shows how the hybrid model is far better than individual ones.

As explain here, Mask R-CNN and YOLOv9 have shown the highest mAP scores in detecting and classifying food items on plates. On the other hand, YOLOv9 had much faster inference speed compared to Mask R-CNN, which is more favorable for real-time applications. While Hybrid approach gives far better performance than these two.

Table 2. Sample results of Food Detection using Mask R-CNN

Modle	mAP (%)	Precision (%)	Recall (%)	F1-Score (%)
YOLOv9	89.50	87.30	86.15	85.93
Mask R-CNN	81.15	80.43	81.50	80.78
Mask R-CNN+ YOLOv9(Hybrid)	91.57	91.28	90.34	90.17

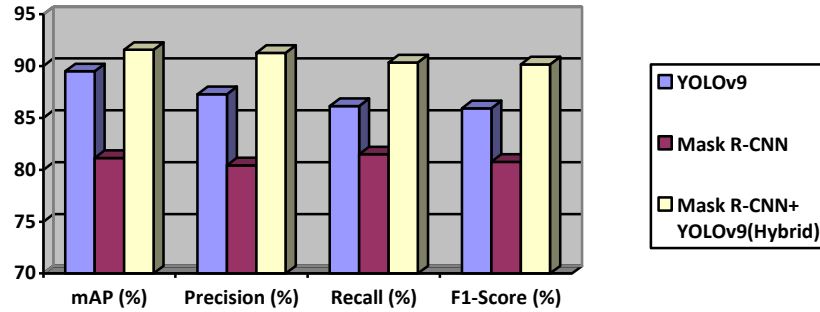


Fig. 4. Statical Graph of result.

In Table 2. And Fig. 3. the output value of YOLOv9, RCNN, and Hybrid model are given. YOLOv9 is the fastest, scoring 89.50% in mean Average Precision (mAP), while RCNN scores 81.15% due to its accurate segmentation. The hybrid model performs even better, with a score of 91.57% mAP and a 90.17% F1 score which is far better than others, successfully balancing speed and accuracy in detection and segmentation.

Table 3. Class-wise Performance for Food sample Detection (using YOLOv9 + Mask R-CNN)

Class	YOLOv9 (mAP %)	MASK R-CNN (mAP %)	Hybrid (YOLOv9+Mask R-CNN)
Rice	73.50	76.00	82.30
Alooparatha	68.30	74.20	80.90
Rasgulla	65.10	67.00	75.50
Papad	87.78	85.60	90.78
Roti	83.17	80.90	86.57

Sample results of food detection using Mask R-CNN and YOLOv9 are shown in Table 3. The ability for both models to detect different types of food including those with complex shapes and occlusions was established. While Mask R-CNN produced better instance segmentation masks, YOLOv9 had faster inference times. ADD HYBRID LINESS

5 Conclusion and Future Work

The hybrid model combines YOLOv9 and RCNN. YOLOv9 provides fast and accurate object detection. RCNN offers precise segmentation. This combination shows great potential for improving object detection tasks. The hybrid approach uses YOLOv9's real-

time detection and RCNN's pixel-level accuracy. This synergy leads to strong performance on complex datasets, especially for food sample detection. The results highlight the usefulness of hybrid models. They can overcome the problem of single methods. This opens the door for more efficient and accurate solutions in various fields. Future research can refine and expand this approach to improve its versatility and scalability.

The model performance can be improved using data augmentation to increase diversity of the training data, transfer learning to leverage a pre-trained model, and changing the architecture to be more accurate and/or efficient. This helps enhance robustness and reduce the training time.

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